

Nguyen Hoang Anh

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Summary

Backend-focused Software Engineer Intern with hands-on experience designing and building REST APIs, AI-integrated pipelines, and cloud-deployed applications across multiple stacks (**Java**, **Python**, **C#**). Experienced in applying **Clean Architecture** to real-world projects. Strong interest in system design, concurrency, and applied AI (RAG, LLM integration).

Technical Skills

Languages: Java, Python(basic), C#

Frameworks & Libraries: Spring Boot, ASP.NET Core, LangChain

Databases: PostgreSQL, MSSQL, Redis, FAISS (Vector DB)

Tools & DevOps: Docker, Git, Postman, Swagger

Testing: JUnit 5, Mockito, Integration Testing

Concepts: RESTful APIs, Clean Architecture, JWT Authentication, OOP, SOLID

Projects

— PROJECT 1: ExamSlot — **Backend Developer** — **ExamSlot — High-Concurrency System**
2025 (In Development)

Team size: 1

- Designed a backend system for high-concurrency exam slot registration, addressing common peak-time overload issues in university portals.
- Implemented a 2-layer concurrency control mechanism using **Redis Distributed Lock** and **Optimistic Locking** (@Version) to maintain slot consistency under concurrent access.
- Applied asynchronous processing with **RabbitMQ**, allowing registration requests to return quickly while handling confirmation and notifications in the background.
- Built a real-time **Watchlist** feature using **WebSocket**, enabling multiple users to receive simultaneous slot availability updates.

Tech: Java 17, Spring Boot 3, PostgreSQL, Redis, RabbitMQ, WebSocket, Spring Security + JWT, JUnit 5, Docker

SmartDoc AI — Offline Document Q&A System

Feb.2026-Presnet

Team size: 4

- Led a team of 4 to build an offline document Q&A system (PDF, DOCX) running on local hardware via Ollama, ensuring data privacy without Internet dependency.
- Designed and implemented a modular **RAG pipeline** using LangChain and FAISS for document chunking, embedding, and retrieval.
- Improved retrieval quality by applying **Cross-Encoder re-ranking** and experimented with query rewriting and response validation techniques.
- Explored **Graph-based retrieval (GraphRAG)** and conducted basic comparisons with standard RAG to evaluate potential improvements.

Tech: Python, LangChain, Ollama, Qwen2.5, Streamlit

Backend Developer — InteractHub — Social Media Platform Backend

Feb.2026-Present

Team size: 4

- Designed system architecture using **Clean Architecture + Onion pattern** (API / Core / Infrastructure / Tests layers) — enabling clear separation of concerns and independent testability.
- Implemented **JWT-based authentication** with ASP.NET Core Identity, covering full auth flows: register, login, logout, forgot password, and reset password.
- Built core modules: Posts, File Upload with **Azure Blob Storage**, and Notifications REST API; developing real-time layer using **SignalR**.
- Deployed backend to **Azure App Service** with **Azure SQL Database** and authored full team documentation (README, API reference, local setup guide).

Tech: C# ASP.NET Core 8, EF Core, MSSQL, JWT, SignalR, Azure Blob Storage, Azure App Service, Swagger

Education

Sai Gon University

2023 – Expected 2028

Bachelor of Software Engineering